FINAL EXPLANATION: MATHEMATICS GRADE 10

Student Assessment Document

FINAL EXPLANATION: MATHEMATICS GRADE 10 – SUB-STRAND 4.2: PROBABILITY I	
Student Assessment Document	
Student Name	____
Class	____
Date	____

INSTRUCTIONS FOR STUDENTS

This is your Final Explanation assessment. You will write a structured response to the driving question using everything you have learned across the 9 lessons in this unit. Read the driving question carefully, then complete each section below in your own words. Use mathematics, vocabulary from the unit, and evidence from Mama Njeri's mandazi data to support every claim you make. Your explanation should show that you can connect probability concepts to real-life decision-making — not just define terms. Write clearly and in full sentences. A strong response does not just recall facts; it reasons, justifies, and applies.

Section 1: Setting Up the Sample Space

Before you can measure likelihood, you need to know what outcomes are possible. Describe Mama Njeri's situation by defining the sample space for her daily mandazi sales. What counts as a 'good day'? What counts as a 'bad day'? Use correct probability vocabulary (experiment, trial, outcome, event, sample space) and explain what each term means in the context of her stall at Gikomba Market.	Mama Njeri's situation is a probability experiment because the outcome — how many mandazi she sells each day — is uncertain, even though the process (frying and selling mandazi) is repeated in a similar way every day. Each day she opens her stall is one trial. An outcome is the specific number of mandazi sold on that day. The sample space (S) is the set of all possible outcomes, which in her case could be any whole number from 0 (she sells nothing) up to however many mandazi she fried that morning — for example, if she fries 50 mandazi, then S = {0, 1, 2, 3, …, 50}. An event is a subset of the sample space we are interested in. The event 'good day' (let's call it event G) is the set of outcomes where she sells at least 80% of her stock — so if she fries 50 mandazi, a good day means selling 40 or more, giving G = {40, 41, 42, …, 50}. A 'bad day' is the complementary event G', meaning she sells fewer than 40 mandazi. Defining the sample space precisely is the first step, because without knowing all possible outcomes, we cannot assign probabilities correctly.

Section 2: Measuring Probability — Theoretical and Experimental

Mama Njeri has kept a notebook of her sales for three months	There are two main ways to calculate probability. Theoretical probability is used when all outcomes are equally likely and we can work out the probability using logic alone. It is defined as: P(event) = Number of favourable outcomes ÷ Total number of outcomes

(about 90 days). Explain TWO ways probability can be calculated: (a) theoretical probability and (b) experimental (relative frequency) probability. Which type is more useful for Mama Njeri's problem, and why? Use a worked numerical example with realistic numbers from her notebook data to support your answer.	in the sample space. For example, if Mama Njeri were rolling a fair die to decide her batch size, the probability of rolling a 6 would be 1/6 ≈ 0.167, because each face is equally likely. However, mandazi sales do not have equally likely outcomes — selling 50 is not just as likely as selling 10 — so pure theoretical probability based on equally likely outcomes does not fully apply here. Experimental probability (relative frequency) is calculated from real data: P(event) ≈ Number of times the event occurred ÷ Total number of trials. This is more useful for Mama Njeri. Suppose her notebook shows that out of 90 days, she had 54 good days (selling ≥ 80% of her stock). Then: P(good day) ≈ 54 ÷ 90 = 0.6 = 60%. This means that based on her past records, roughly 6 out of every 10 days are good days. The more trials she records, the more reliable this estimate becomes — which is why her notebook of 90 days gives a much better estimate than a notebook of only 5 days would.

Section 3: The Law of Large Numbers — Why the Fraction Stabilises	
Students examining Mama Njeri's notebook noticed something surprising: even though individual daily sales jump around unpredictably, the proportion of 'good days' seems to settle towards a stable value the longer they look. Explain this observation using the Law of Large Numbers. Why are single days unpredictable, yet long-run patterns reliable? Use a sketch, table, or numerical example to illustrate how relative frequency changes as the number of trials increases.	The key idea here is the Law of Large Numbers: as the number of trials increases, the experimental probability (relative frequency) gets closer and closer to the true probability of the event. This is why Mama Njeri's proportion of good days stabilises over time, even though she cannot predict any single day. Think of it this way — imagine flipping a fair coin. On the first 5 flips you might get 4 heads (relative frequency = 0.8), which seems far from 0.5. But after 1,000 flips, the relative frequency of heads will be very close to 0.5. The same principle applies to Mama Njeri's sales. After just 5 days, her proportion of good days might be 1/5 = 0.2 or 4/5 = 0.8 — these early values are unreliable. But after 90 days, the proportion has had many more trials to 'average out' the extreme days, and it converges towards the true underlying probability. A table helps show this: after 10 days — 7 good days — relative frequency = 0.70; after 30 days — 19 good days — relative frequency = 0.63; after 60 days — 37 good days — relative frequency = 0.62; after 90 days — 54 good days — relative frequency = 0.60. The values are settling towards 0.60. Individual days are unpredictable because each one depends on many uncontrolled factors (weather, nearby competition, whether it is a market day). But across many days, these random factors balance out, revealing the underlying pattern. This is the core tension of the unit: uncertainty in the short run, predictability in the long run.

Section 4: Using Probability to Make a Smarter Decision	
Now apply your understanding directly to Mama Njeri's problem. She wants to decide how many mandazi to fry each morning. Using the probability you calculated from her notebook, explain the decision she should make and why. Consider at least TWO outcomes or scenarios (e.g.,	Using Mama Njeri's experimental probability of a good day as P(good day) = 0.60, we can help her make a smarter decision. Suppose she is choosing between frying 40 mandazi or 50 mandazi each morning, and each mandazi costs her KSh 5 to make and sells for KSh 15. Scenario A — She fries 50 mandazi: On a good day (probability 0.60), she sells at least 40 (≥ 80%), earning approximately KSh 600 revenue. On a bad day (probability 0.40), she sells fewer than 40 — say on average 30 — and wastes 20 unsold mandazi, losing KSh 100 in wasted ingredients. Scenario B — She fries 40 mandazi: On a good day, she sells all 40 and earns KSh 600 with no waste, but she also misses the chance to earn more from the extra customers she turns away. On a bad day (probability 0.40), she sells about 30 and wastes only 10 mandazi, losing only KSh 50. Probability tells Mama Njeri that a bad day happens 40% of the time — that is 2 days out of every 5. This is frequent enough that consistently frying 50 mandazi is quite risky. A probability-informed decision might be to fry 40–45 mandazi on regular weekdays, and increase to 50 on Saturdays and public market days if her data shows those days have a higher probability of being good days. The key insight is that probability

frying too many vs. frying too few) and connect them to probability. Show any calculations clearly.	does not eliminate uncertainty — it gives you a rational, evidence-based way to manage it and reduce unnecessary losses.

Section 5: Answering the Driving Question	
Bring everything together. In a well-reasoned paragraph (or two), answer the driving question directly: How can we use mathematics to measure and predict the likelihood of uncertain everyday events — like Mama Njeri's mandazi sales — so we can make smarter decisions even when we cannot be certain of the outcome? Your answer must reference: the concept of probability, experimental vs. theoretical probability, the Law of Large Numbers, and at least one real-life connection beyond Mama Njeri's stall.	Mathematics gives us a powerful tool — probability — to make sense of uncertainty. Even though no one can know exactly how many mandazi Mama Njeri will sell tomorrow, we can use her three months of notebook data to calculate the experimental probability of a good day (approximately 0.60, or 60%). This number is not a guess; it is a mathematically grounded estimate built from 90 real trials. The Law of Large Numbers assures us that the more data Mama Njeri collects, the more reliable this estimate becomes — so keeping her notebook is itself a mathematical act. With this probability in hand, she can weigh her options, anticipate likely losses, and choose a batch size that balances risk and reward. She cannot be certain, but she can be informed. This same logic extends far beyond a mandazi stall. Kenyan farmers in the Rift Valley use historical rainfall data to estimate the probability of a good harvest before deciding how many hectares to plant. Doctors use probability to explain to patients the likelihood that a treatment will work. Insurance companies, sports coaches, and even Kenya Power when managing electricity demand — all of them use probability to make better decisions in the face of uncertainty. The fundamental lesson of this unit is that individual outcomes are unpredictable, but mathematical patterns across many outcomes are reliable — and that reliability is what makes probability one of the most useful branches of mathematics in everyday life.

RUBRIC			
Criterion	**Excellent (4)**	**Proficient (3)**	**Developing (1–2)**
Probability Concepts & Vocabulary	Accurately defines and consistently uses all key terms (experiment, trial, outcome, event, sample space, theoretical probability, experimental probability) in the correct context throughout the explanation. Definitions are precise and connected to Mama Njeri's scenario.	Correctly uses most key probability terms with minor errors or omissions. Definitions are mostly accurate and generally connected to the scenario.	Uses some probability vocabulary but with notable inaccuracies, confusion between terms (e.g., outcome vs. event), or terms are defined in isolation without connecting to the scenario.
Mathematical Accuracy & Calculations	All probability calculations are correct and clearly set out (e.g., P(event) = favourable outcomes ÷ total outcomes or relative frequency formula). Working is shown step by step. Numbers used are realistic and consistent across sections.	Calculations are mostly correct with one or two minor arithmetic or formula errors that do not undermine the overall argument. Working is partially shown.	Calculations contain significant errors, the wrong formula is applied, or numerical examples are missing, vague, or inconsistent across sections.

Law of Large Numbers & Long-Run Reasoning	Clearly explains why individual daily outcomes are unpredictable yet long-run proportions are stable. Explicitly connects this to the Law of Large Numbers with a table, sketch, or numerical sequence showing relative frequency converging over increasing trials.	Explains the stabilising pattern and references the Law of Large Numbers, but the illustration (table/sketch/numbers) is incomplete or only partially developed.	Mentions that the fraction stabilises but does not explain why, or confuses short-run variability with long-run stability without referencing the Law of Large Numbers.
Application to Real-Life Decision-Making	Constructs a clear, logical argument for how Mama Njeri (or another real-life actor) should use probability to make a decision. Considers at least two scenarios, links each to its probability, and draws a justified recommendation. At least one additional real-life context beyond the mandazi stall is referenced.	Applies probability to Mama Njeri's decision with a reasonable argument and some scenario comparison. The recommendation is present but may lack full justification or a second real-life context.	Attempts to connect probability to decision-making but the argument is vague, no scenarios are compared, or the recommendation is not supported by probability values.
Coherence & Quality of Final Answer	The final answer directly and fully addresses the driving question, synthesising all four concepts (probability, experimental vs. theoretical, Law of Large Numbers, real-life application) into a cohesive, well-reasoned response. Writing is clear, logical, and student-voice appropriate.	The final answer addresses the driving question and touches on most required concepts, but the synthesis is partial — some ideas are listed rather than integrated into a flowing argument.	The final answer is present but only partially addresses the driving question, omits one or more required concepts, or reads as a collection of disconnected statements rather than a coherent explanation.